

IDENTITY-INDEXED LOCAL TEMPO ROUTING FOR MULTI-PERSON REPETITION COUNTING

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ABSTRACT

Multi-person repetition counting estimates a count for each tracked person rather than only a scene total. The task is difficult when people move asynchronously, individual pace changes within a trajectory, and missing poses turn each trajectory into a masked nonstationary signal. We propose an identity-indexed signal-processing specification that receives supplied pose tracks and targets counting without person-wise count or period labels, while acknowledging external supervision in the pose and tracking frontends. Overlapping windows share an encoder and fast, medium, and slow response experts; masked autocorrelation and Fourier evidence yield local period cues and confidence. Soft routing mixes expert responses, and normalized overlap-add reconstructs one response and decodes it once per identity. This design separates per-track state from shared parameters and avoids independently counting every overlapping window, removing one structural source of boundary double-counting. **Experimental evidence is pending synchronization with the collaborator’s frozen implementation and audited MultiRep runs.**

Index Terms— repetition counting, multi-person signals, local tempo, pose trajectories, temporal routing

1. INTRODUCTION

Person-wise repetition counts matter whenever several people exercise, rehabilitate, train, or perform repetitive work in the same scene. A single scene total cannot say who completed each repetition, paused, or changed pace. We therefore study multi-person repetition counting from supplied identity-indexed pose tracks. For N tracks, the primary output is a vector of N counts; their sum is only a diagnostic. The signal is difficult because people move asynchronously, a person’s period may change inside one track, and occlusion creates irregular missing intervals.

Repetition counting progressed from hand-designed periodic motion analysis [1, 2] to learned temporal similarity and correlation models such as RepNet and TransRAC [3, 4]. PoseRAC emphasizes salient poses [5], while PAMS learns period-adaptive multi-scale consistency from pose sequences without count labels in its counting objective [6]. MultiCounter establishes multi-person counting in untrimmed video [7]; MultiCounter+ further addresses identity continuity and long-versus-short repetition regimes [8]. These advances cover multi-person outputs, pose-driven counting, low-label objectives, and broad period adaptation. They do not, however, resolve the narrower signal problem considered here: routing each overlapping window of each supplied identity track according to tempo that changes within that track, then reconstructing one response before count decoding. Recent counters also explore dynamic queries and

decoupled spatiotemporal attention [9, 10]; those architectural directions are complementary to the identity-local temporal scale selection studied here. Windowed spectral estimation, frequency analysis under irregular sampling, overlap-add reconstruction, and learned expert routing are established ideas [11–14]; no one of them is claimed as novel here. Our narrower, result-free hypothesis is that identity-local tempo routing plus reconstruct-before-decode is more robust to within-track tempo changes than a parameter-matched global-tempo counter while preserving behavior on stationary tracks.

The proposed formulation separates inter-person asynchrony from intra-person nonstationarity. Each pose track is a masked multidimensional signal with identity-indexed state, whereas learned encoder and expert parameters are shared across people. The intended counting objective uses neither person-wise count labels nor person-wise period labels, but this qualifier does not extend to the perception stack: pose estimators and trackers may use externally supervised training data. The exact supervision path, including model selection and calibration, remains **SYNC-REQUIRED: supervision firewall** until the collaborator implementation is frozen. Selecting a global long-versus-short regime differs from allowing several pace changes inside one identity trajectory. Mask-aware local estimation also exposes missing observations instead of conflating invalid frames with stationary motion. These distinctions drive both the reconstruction invariant and the planned diagnostic.

Our signal path follows a causal order. A shared encoder represents overlapping masked windows. Local autocorrelation and Fourier evidence estimate period and confidence for each identity-window pair. A soft router mixes fast, medium, and slow shared response experts; normalized overlap-add then reconstructs one continuous response per identity, and one trajectory-level decoder processes it once. Figure 1 shows this specification and its perception boundary. The local worktree does not yet contain the multi-person implementation, so router, expert, reconstruction, and decoder details remain proposal-level interfaces rather than measured components.

The pre-results manuscript makes three falsifiable contributions:

- It formulates person-wise counting as inference on identity-indexed, masked, nonstationary pose signals, with scene totals kept diagnostic.
- It specifies window-local tempo evidence, soft routing over shared response experts, normalized reconstruction, and one decode per track; every source-sensitive interface is explicitly marked for synchronization.
- It specifies planned protocol-separated comparisons, paired mechanism ablations, and a piecewise time-warp diagnostic. Empirical claims remain disabled until frozen MultiRep artifacts pass the evidence gates.

2. IDENTITY-INDEXED LOCAL TEMPO ROUTING

The counter begins after perception: it receives pose trajectories with supplied identity indices and validity masks. Oracle tracks, tracks from a frozen common frontend, and native method pipelines define different experimental protocols and are never ranked together. Parameters are shared, whereas buffers, periodic evidence, masks, and responses remain identity-indexed; no window combines people. Figure 1 summarizes the proposed specification, whose tensor layouts and learned modules remain **SYNC-REQUIRED: collaborator source binding**. For predicted tracks, an index denotes a current tracklet, not a guaranteed person. Fragmentation, re-entry, switches, retirement, and any reconciliation belong to the **SYNC-REQUIRED: track lifecycle contract**.

2.1. Masked identity-indexed representation

For person $i \in \{1, \dots, N\}$, let $\mathbf{X}_i$ contain J joints with C coordinates over T_i frames, and let $m_i(t)$ mark a valid observation. Overlapping supports $\mathcal{W}_{i\ell}$ select local windows. A shared encoder $\mathcal{E}_\theta$ maps the masked samples to features, while the primary output remains a person-wise count vector:

$$\mathbf{X}_i \in \mathbb{R}^{T_i \times J \times C}, \quad m_i \in \{0, 1\}^{T_i}, \quad \hat{\mathbf{c}} = [\hat{c}_1, \dots, \hat{c}_N]^\top \quad (1)$$

$$\mathbf{z}_{i\ell}(t) = \mathcal{E}_\theta(\mathbf{X}_i[\mathcal{W}_{i\ell}], m_i[\mathcal{W}_{i\ell}])_t, \quad t \in \mathcal{W}_{i\ell}$$

The shared θ does not mix identity state. Window geometry, pose normalization, padding, short tracks, and buffer lifecycle are **SYNC-REQUIRED: track tensor and lifecycle contract**. Masks accompany every transformation, and windows never cross identity timelines. A scene total $\sum_i \hat{c}_i$ is diagnostic. Permutation, isolated-mask, and short-track tests will check these invariants once the source interface exists.

2.2. Local masked periodic evidence

Tempo must vary with the window rather than being fixed for the whole track. Let $s_{i\ell}(t)$ denote the scalar evidence signal provisionally projected from $\mathbf{z}_{i\ell}(t)$, and $q_{i\ell}(t)$ its windowed validity mask. We define a mask-normalized non-circular autocorrelation and a normalized Fourier score, then map both to a period estimate and confidence:

$$\bar{s}_{i\ell} = \frac{\sum_t q_{i\ell}(t) s_{i\ell}(t)}{\sum_t q_{i\ell}(t) + \epsilon}$$

$$A_{i\ell}(\tau) = \frac{\sum_t q_{i\ell}(t) q_{i\ell}(t + \tau) [s_{i\ell}(t) - \bar{s}_{i\ell}] [s_{i\ell}(t + \tau) - \bar{s}_{i\ell}]}{\sum_t q_{i\ell}(t) q_{i\ell}(t + \tau) + \epsilon}$$

$$F_{i\ell}(\omega) = \frac{|\sum_t q_{i\ell}(t) [s_{i\ell}(t) - \bar{s}_{i\ell}] e^{-j\omega t}|^2}{\max_{\omega'} |\sum_t q_{i\ell}(t) [s_{i\ell}(t) - \bar{s}_{i\ell}] e^{-j\omega' t}|^2 + \epsilon}$$

$$(\hat{p}_{i\ell}, \gamma_{i\ell}) = \Phi(\{A_{i\ell}(\tau)\}_{\tau \in \mathcal{T}}, \{F_{i\ell}(\omega)\}_{\omega \in \Omega}) \quad (2)$$

Indices (i, ℓ) expose within-track drift. Projection, centering, domains and units, interpolation, confidence, and fallback are **SYNC-REQUIRED: period estimator**; Eq. 2 fixes only the semantics of local masked evidence. Confidence must measure available evidence rather than a post-routing label, and low-support windows must be masked or use one frozen fallback in training and evaluation. Since periodic visibility can induce spectral peaks, the implementation must set out-of-range masks and minimum lag-pair support and pass periodic/nonperiodic missingness controls. It must also choose pair-dependent centering or another irregular-sample estimator [12].

2.3. Soft routing over shared response experts

Let $\mathcal{K} = \{\text{fast, medium, slow}\}$ index shared experts. A proposed router consumes the window representation, local period, and confidence, and returns simplex weights. Each expert emits a continuous local response rather than an independently rounded count:

$$\mathbf{g}_{i\ell} = \text{softmax}(\rho_\psi(\mathbf{z}_{i\ell}, \hat{p}_{i\ell}, \gamma_{i\ell}) / \tau_r), \quad g_{i\ell k} \geq 0, \quad \sum_{k \in \mathcal{K}} g_{i\ell k} = 1$$

$$\mathbf{r}_{i\ell}^{(k)} = \mathcal{H}_{\phi_k}^{(k)}(\mathbf{z}_{i\ell}), \quad k \in \mathcal{K} \quad (3)$$

Soft weights retain uncertainty at tempo transitions. Router inputs, temperature, expert architecture, gradients, sharing, and fallback remain **SYNC-REQUIRED: router and expert implementation**. Fast/medium/slow are intended scales, not observed specializations: paired controls must test expert use, collapse, frozen tempo bands, and hard/uniform routing. Shared experts keep parameter count independent of the number of tracks, but no runtime or memory claim is enabled without profiling.

2.4. Response reconstruction and one track decode

The final stage mixes expert responses before merging overlapping windows. Let $a_{i\ell}(t)$ be a taper on window ℓ , and let $q_{i\ell}(t)$ exclude invalid samples. Normalized overlap-add forms a single track response R_i , after which one decoder extracts events:

$$\tilde{r}_{i\ell}(t) = \sum_{k \in \mathcal{K}} g_{i\ell k} r_{i\ell}^{(k)}(t)$$

$$R_i(t) = \frac{\sum_{\ell: t \in \mathcal{W}_{i\ell}} a_{i\ell}(t) q_{i\ell}(t) \tilde{r}_{i\ell}(t)}{\sum_{\ell: t \in \mathcal{W}_{i\ell}} a_{i\ell}(t) q_{i\ell}(t) + \epsilon} \quad (4)$$

$$\hat{\mathcal{E}}_i = \mathcal{D}(R_i, m_i), \quad \hat{c}_i = |\hat{\mathcal{E}}_i|$$

This order avoids counting every window independently; it does not guarantee one peak. Decoded samples require aligned timestamps and coverage $\sum_\ell a_{i\ell}(t) q_{i\ell}(t) > 0$. Taper, gaps, event confidence, peak parameters, and decoder semantics remain **SYNC-REQUIRED: reconstruction and decoder**. PAMS-TCC is the documented starting objective [6]; $\mathcal{L}_{\text{PAMS-TCC}} + \lambda_r \mathcal{L}_{\text{route}} + \lambda_t \mathcal{L}_{\text{track}}$ is only a synchronization contract. The latter terms, their existence, formulas, weights, samples, supervision, and gradient paths must bind to code; absent terms will be deleted. Overlap-add preserves continuous evidence before one masked trajectory-level decode. A boundary test will place an event at several offsets and test one-time decoding.

The supervision audit must inspect loaders, pseudo-targets, auxiliary losses, selection, stopping, calibration, and inference fallbacks at one commit. Until it passes, the paper states only the intended absence of person-wise count and period labels in the counting objective and discloses perception supervision.

3. PLANNED EXPERIMENTS AND ANALYSIS

The protocol is planned but is neither frozen nor executed for the proposed multi-person system. Existing single-person development artifacts are excluded from every table because they neither implement the proposed method nor use the target benchmark. Result prose will be enabled only after a frozen collaborator bundle passes experiment and claim audits.

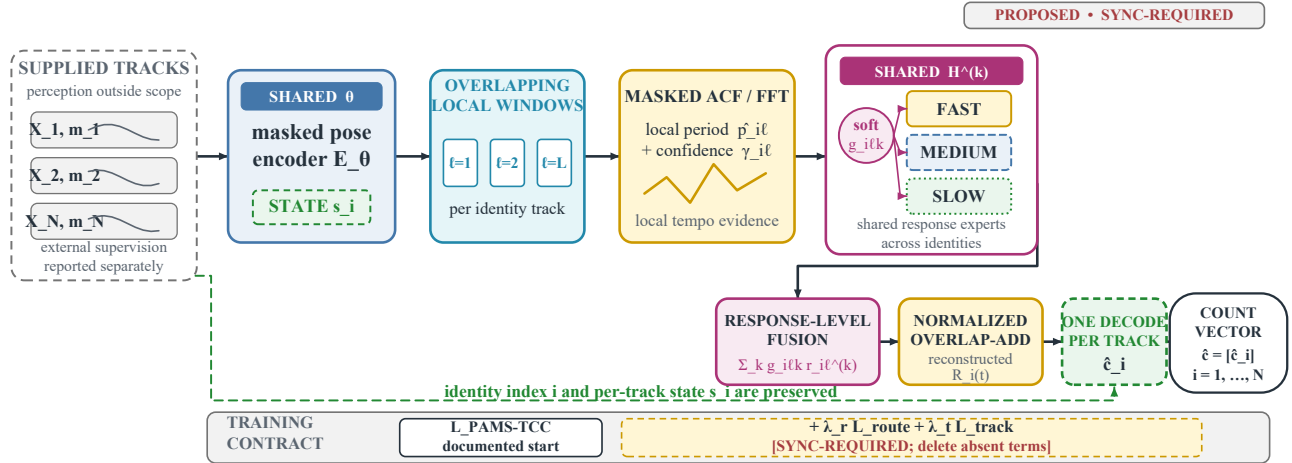


Fig. 1. Proposed signal path, not an implementation or result claim. Supplied pose tracks and masks retain identity-indexed state while sharing encoder and expert parameters. Each window yields local periodic evidence, soft routing mixes continuous expert responses, normalized overlap-add reconstructs one track response, and one decoder returns each person’s count. Red synchronization markers identify interfaces that must be replaced by source-bound definitions.

3.1. Protocol and metrics

The result manifest must bind the MultiRep release, official split, video and annotation checksums, evaluator revision, preprocessing fingerprint, pose modality, supervision path, hardware, seeds, configuration, and raw logs. We separate three protocols: oracle tracks, tracks from one frozen common frontend, and each method’s native pipeline. Native rows provide context but cannot form a matched ranking against common-track rows. Predicted-track evaluations retain HOTA [15], IDF1, and identity switches [16] as frontend diagnostics, separate from the counting metrics. Their proposed-method values remain **PENDING**, **PENDING**, and **PENDING**, respectively, until the same frozen bundle binds them. For every predicted-track protocol, the manifest must also freeze tracklet creation, reset, re-entry, fragmentation, merge, retirement, and reconciliation rules, plus the assignment from predicted tracklets to ground-truth people. Until these semantics are source-bound, predicted-track count rows stay disabled.

The principal counting metrics are Period-mAP, AP50, AP75, AvgMAE, and AvgOBO. Every reproduced learned method requires at least three named seeds frozen before results are inspected; we will report the mean, sample standard deviation, and a paired video-cluster bootstrap 95% confidence interval. A published checkpoint may appear as a contextual row, but it cannot support a variance estimate. Dataset and evaluator fields remain **PENDING** and **PENDING**. The frozen bundle must also contain per-person ground truth, per-person predictions, track identifiers, evaluator outputs, and the exact commands that produced each row. Checksums bind the structured source of every table cell, so the manuscript cannot silently mix releases, frontends, or reruns.

3.2. Main comparison

Table 1 separates native MultiCounter variants from a common-track family. The latter uses one identical tracking frontend for Track-PAMS, a track-global tempo control, and the proposed local router. RepNet and TransRAC [3, 4], PoseRAC [5], dynamic-query counting [9], and dual-branch spatiotemporal counting [10] enter this block only if input, output, tracking, and evaluation parity can be demon-

DATA-PENDING DIAGNOSTIC SHELL

local period | routing probability/entropy
reconstructed response | clean-to-corrupt change

Fig. 2. Planned nonstationary-tempo diagnostic; the box contains no measured data. A per-person piecewise time-warp will align local period, routing probabilities or entropy, the reconstructed response, and paired clean-to-corrupt changes under a frozen selection rule.

strated. No cell will be emphasized across incompatible protocols. The table will be generated from frozen structured results rather than manually transcribed values. A baseline enters the common-track block only if its temporal input and output can be adapted without privileged labels or method-specific tracking. When such parity cannot be established, the method remains contextual and its unavailable metrics are reported as not applicable rather than imputed.

3.3. Mechanism ablation

Table 2 separates local ACF and Fourier evidence, local against global tempo, single-expert and hard against soft routing, and boundary reconstruction. Every retained row must match parameter count, temporal context, frontend, training budget, paired videos, and seeds, and must point to one commit and configuration. Removal rows for routing-consistency, track-continuity, or fallback behavior will appear only if those components exist in the frozen training graph. This rule prevents a writing placeholder from becoming a simulated ablation. Each contrast will report the paired video-cluster confidence interval as well as its seed aggregate, because an unpaired comparison could confound the component change with video difficulty or initialization. The primary outcome, contrast direction, bootstrap replicates, multiplicity policy, and failure/exclusion rules must be frozen before results are inspected.

Table 1. Planned main comparison. All cells are evidence-controlled; native/contextual and common-track blocks are not ranked across their divider. Arrows specify metric direction, not a performance claim.

Method	Protocol	Period-mAP $\uparrow$	AP50 $\uparrow$	AP75 $\uparrow$	AvgMAE $\downarrow$	AvgOBO $\uparrow$
MultiCounter [7]	native/contextual	PENDING	PENDING	PENDING	PENDING	PENDING
MultiCounter+ [8]	native/contextual	PENDING	PENDING	PENDING	PENDING	PENDING
Track-PAMS [6]	common predicted tracks	PENDING	PENDING	PENDING	PENDING	PENDING
Track-global tempo	common predicted tracks	PENDING	PENDING	PENDING	PENDING	PENDING
Identity-indexed local routing	common predicted tracks	PENDING	PENDING	PENDING	PENDING	PENDING

Table 2. Planned paired mechanism ablations. Deltas are variant minus the full local/soft/OLA row, so positive Δ MAE and negative Δ OBO are worse. Only source-bound variants remain in the final table.

Tempo	Evidence	Router	Fusion	Δ MAE	Δ OBO
global	ACF+FFT	uniform	OLA	PENDING	PENDING
local	ACF only	soft	OLA	PENDING	PENDING
local	FFT only	soft	OLA	PENDING	PENDING
local	ACF+FFT	single	OLA	PENDING	PENDING
local	ACF+FFT	hard	OLA	PENDING	PENDING
local	ACF+FFT	soft	average	PENDING	PENDING
local	ACF+FFT	soft	OLA	PENDING	PENDING

3.4. Nonstationary-tempo diagnostic

Figure 2 defines the planned mechanism test. A frozen piecewise time-warp will change one person’s pace while preserving the evaluation unit and paired identity. The final plot will use either a selection-rule-bound example or an aggregate fixed before inspection. It must record source data, generator hash, warp schedule, routing entropy, reconstructed response, and clean-to-corrupt metric change. Counts by number of people, occlusion level, and identity-switch severity remain secondary diagnostics. The figure can support a mechanism statement only after the paired artifact and its generation trace pass audit.

Separate mask-leakage controls will hold motion fixed while applying periodic and nonperiodic missingness, and will hold the mask fixed while changing tempo. They test whether period confidence or routing succeeds by reading visibility rather than motion. The generator, missingness schedules, minimum support, and all exclusion rules must be frozen before any output is inspected.

The corruption changes tempo inside a track without changing its identity or the number of source events. Clean and warped samples therefore share an evaluation cluster, allowing paired changes to isolate the intended stressor. Occlusion and identity-switch stress tests are generated separately so that a tempo conclusion is not attributed to a simultaneous perception failure. Any representative visualization follows a frozen selection rule; otherwise, the paper reports an aggregate to avoid post hoc example choice.

4. CONCLUSION

We formulated multi-person repetition counting as inference on masked, identity-indexed pose signals whose tempo can change within a track. The proposed specification estimates local periodic evidence, softly routes shared response experts, reconstructs each identity response with normalized overlap-add, and decodes that response once. This pre-results draft has no local multi-person implementation or eligible MultiRep results, and its pose and tracking frontends rely on external supervision. Completion requires a frozen collaborator commit, source-bound losses and interfaces, matched oracle and common-track evaluation, and paired piecewise-tempo

diagnostics. Empirical and citation audits must then be rerun before any quantitative or comparative conclusion is enabled.

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